



Cairo University
Faculty of Economics and Political Science
Statistics department
4th Level / Arabic section



Survival Analysis and Mortality Risk Factors in Primary Biliary Cirrhosis

Prepared by

Nadin Aladdin Mohamed
(5220143)

Akram Wael Mohamed
(5220650)

Rana Adel Mostafa
(5220376)

Mostafa Hesham Sarhan
(5220440)

Roaa Mostafa Saad
(5220525)

Supervised by

Dr. Wafaa Ibrahim

1. Introduction

Survival analysis is a fundamental branch of biostatistics concerned with modelling the time until a defined event of interest occurs. Unlike standard regression, survival methods explicitly account for censored observations – patients who leave the study before the event occurs making them uniquely suited to longitudinal clinical data.

Primary Biliary Cirrhosis is a chronic, progressive autoimmune liver disease characterized by the destruction of intrahepatic bile ducts, ultimately leading to cirrhosis, liver failure, and death if untreated. PBC predominantly affects middle-aged women and is associated with a range of clinical and biochemical abnormalities including elevated serum bilirubin, decreased albumin, elevated copper, and prolonged prothrombin time.

This project performs a complete survival analysis of the Mayo Clinic PBC randomized clinical trial dataset. The primary objectives are:

- estimate overall survival using the Kaplan–Meier method.
- compare survival across clinically relevant patient subgroups.
- identify independent prognostic predictors of mortality using a multivariable Cox proportional hazards regression model.
- assess and address violations of the proportional hazards assumption to ensure the validity and reliability of the final model.

2. Data Description

2.1 Dataset overview

The dataset is the Mayo Clinic Primary Biliary Cirrhosis (PBC) dataset, a well-established benchmark dataset in survival analysis research. It originates from a double-blind, randomized controlled clinical trial conducted between 1974 and 1984 at the Mayo Clinic, evaluating the effect of D-penicillamine versus placebo on survival in patients with PBC.

The dataset contains 418 observations and 20 variables. The primary time variable, originally recorded in days, was converted to years for ease of interpretation. The event/status variable indicates whether the patient died during follow-up (coded as 1 = Dead) or was censored (coded as 0 = Censored). Patients were censored if they were alive at the end of the study, underwent liver transplantation, or were lost to follow-up.

Of the 418 patients, 161 (38.5%) experienced the event of death during the study period, while 257 (61.5%) were censored. The median follow-up time was approximately 4.74 years, with a maximum of 13.13 years.

2.2 Data Dictionary

The table below provides a complete data dictionary for all variables:

Variable	Description	Type	Units/Coding	Role
Years	Follow-up time from enrolment to event/censoring	Continuous	Years	Outcome (time)
Status	Vital status at end of follow-up	Binary	1=Dead, 0=Censored	Outcome (event)
Age	Patient age at enrolment	Continuous	Days	Covariate
Sex	Patient biological sex	Categorical	0=Female, 1=Male	Covariate
Bilirubin	Serum bilirubin concentration	Continuous	mg/dL	Covariate
Albumin	Serum albumin concentration	Continuous	g/dL	Covariate
Edema	Presence and severity of oedema	Ordinal	0/0.5/1	Covariate
Copper	Urine copper level	Continuous	µg/day	Covariate
Prothrombin	Prothrombin time	Continuous	Seconds	Covariate
Stage	Histologic stage of disease	Ordinal	1–4	Covariate
Drug	Treatment arm	Categorical	D-penicillamine / Placebo	Covariate
Cholesterol	Serum cholesterol	Continuous	mg/dL	Covariate
Ascites	Presence of ascites	Binary	N/Y	Covariate
Hepatomegaly	Presence of hepatomegaly	Binary	N/Y	Covariate
Spiders	Presence of spider angiomas	Binary	N/Y	Covariate

2.3 Missing values:

The original dataset contains 1,033 missing values distributed across several variables, notably Cholesterol (n=134 missing), Triglycerides (n=136), Copper (n=108), Alkaline Phosphatase (n=106), and SGOT (n=106). These missing values predominantly affect patients who did not participate in the clinical trial portion (approximately 106 patients). For the primary analyses, only variables with complete data were retained. The complete-case dataset for the multivariable Cox model comprised 310 observations.

3. Descriptive statistics

3.1 Numerical variables

- **Follow-up Time (Years):** The follow-up period ranged from 0.11 to 13.13 years, with a mean of 5.25 years (SD = 3.02) and a median of 4.74 years. The mean being slightly higher than the median indicates a mild right skew, meaning a small number of patients had considerably longer follow-up durations.
- **Age (Days):** Patient age at enrolment ranged from 9,598 to 28,650 days (approximately 26 to 78 years), with a mean of 18,533 days (SD = 3,816) and a median of 18,628 days equivalent to approximately 51 years. The close agreement between mean and median indicates a roughly symmetric age distribution across the cohort.
- **Serum Bilirubin (mg/dL):** Bilirubin showed a strongly right-skewed distribution. The median was 1.40 mg/dL, substantially lower than the mean of 3.22 mg/dL (SD = 4.41), indicating that while most patients had near-normal bilirubin levels, a subset presented with markedly elevated values reaching up to 28.00 mg/dL. Elevated bilirubin is a hallmark of cholestatic liver disease and a key prognostic marker in PBC.
- **Serum Cholesterol (mg/dL):** Cholesterol had 134 missing values (32% of the cohort), the highest proportion of missing data among all variables. Among observed values, the mean was 369.5 mg/dL (SD = 231.9) and the median was 309.5 mg/dL, again reflecting right skew driven by extreme values up to 1,775 mg/dL. Hypercholesterolaemia is common in cholestatic liver conditions.
- **Serum Albumin (g/dL):** Albumin was the most symmetrically distributed biochemical variable, with a mean of 3.50 g/dL and a median of 3.53 g/dL (SD = 0.42). Values ranged from 1.96 to 4.64 g/dL. The normal range for serum albumin is 3.5–5.0 g/dL, indicating that many patients in this cohort had albumin levels at or below the lower limit of normal, consistent with impaired hepatic synthetic function in advanced liver disease.
- **Urinary Copper (µg/day):** Copper levels were right-skewed, with a median of 73.00 µg/day considerably below the mean of 97.65 µg/day (SD = 85.61), and a maximum of 588.00 µg/day. Elevated urinary copper reflects impaired biliary copper excretion, a characteristic feature of cholestatic liver disease. This variable had 108 missing values.
- **Alkaline Phosphatase Alk_Phos (U/L) :** Alkaline phosphatase showed the most extreme right skew of all variables. The median was 1,259.0 U/L while the mean was 1,982.7 U/L (SD =

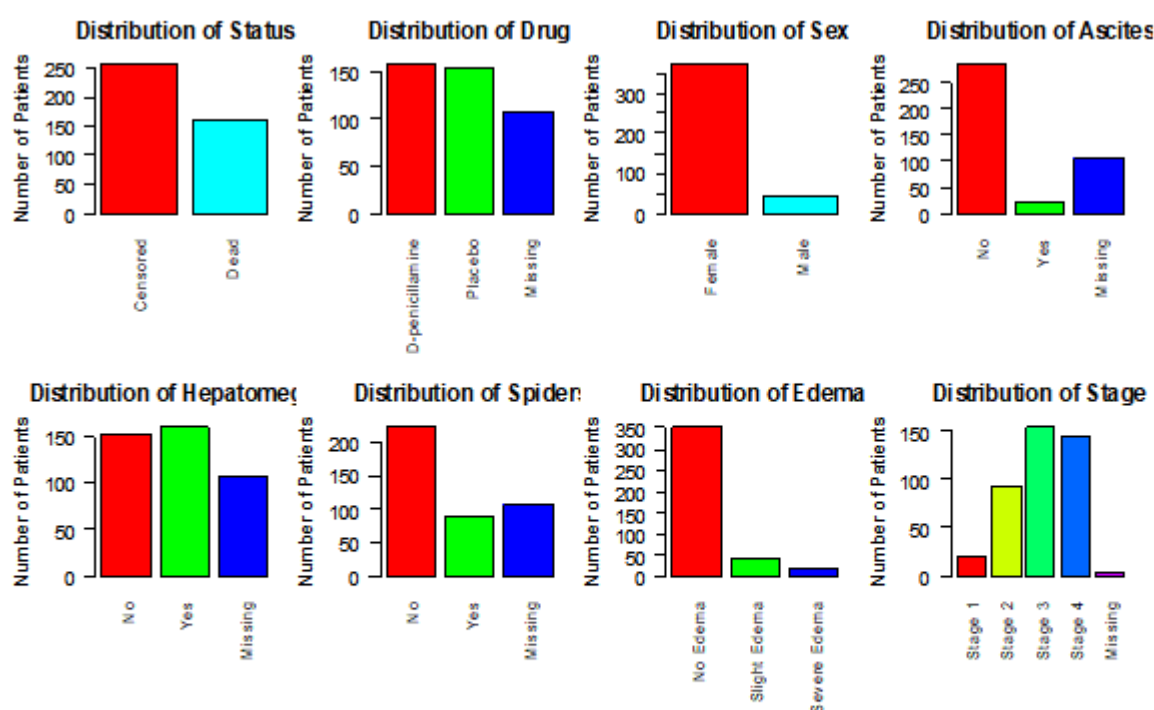
2,140.4), with a maximum of 13,862.4 U/L. This extreme variability and skew reflect the wide spectrum of cholestatic severity across the cohort. Normal values are typically below 120 U/L, confirming that virtually all patients had markedly elevated levels.

- **Serum Glutamic-Oxaloacetic Transaminase (U/L) :**SGOT ranged from 26.35 to 457.25 U/L, with a mean of 122.56 U/L (SD = 56.70) and a median of 114.70 U/L. The moderate right skew indicates that most patients had moderately elevated liver enzyme levels, with a smaller subset showing severe hepatocellular involvement.
- **Triglycerides (mg/dL):** Triglycerides ranged from 33.00 to 598.00 mg/dL, with a mean of 124.70 mg/dL (SD = 65.15) and a median of 108.00 mg/dL. The right skew suggests a minority of patients had markedly elevated triglyceride levels. This variable had the highest number of missing values at 136 observations.
- **Platelets ($\times 10^3/\mu\text{L}$):** Platelet count ranged from 62 to 721 $\times 10^3/\mu\text{L}$, with a mean of 257.0 (SD = 98.33) and a median of 251.0. The distribution was relatively symmetric. Low platelet counts in some patients may reflect portal hypertension and hypersplenism associated with advanced cirrhosis.
- **Prothrombin Time (seconds):** Prothrombin time was the most narrowly distributed variable, ranging from 9.00 to 18.00 seconds with a mean of 10.73 seconds (SD = 1.02) and a median of 10.60 seconds. The normal range is approximately 9–12 seconds, indicating that while most patients had values within or near the normal range, a subset had meaningfully prolonged clotting times reflecting impaired hepatic coagulation factor synthesis. Only 2 values were missing.

3.2 Categorical variables

- **Vital Status (Status) :**418 patients in the cohort, 257 (61.5%) were censored meaning they were alive at the end of the study or lost to follow-up before the event occurred while 161 patients (38.5%) died during the follow-up period. This event rate of approximately 38% is sufficient for reliable survival modelling and Cox regression estimation.
- **Treatment Drug:** The cohort was nearly evenly divided between treatment arms: 158 patients (37.8%) received D-penicillamine and 154 patients (36.8%) received placebo. Notably, 106 patients (25.4%) had no drug assignment recorded, as these individuals were not part of the randomised trial portion of the study and were enrolled as observational cases only.
- **Sex:** The cohort was predominantly female, with 374 female patients (89.5%) compared to only 44 male patients (10.5%). This striking sex imbalance is consistent with the well-established epidemiology of PBC, which disproportionately affects middle-aged women. The small number of male patients limits sex-stratified inferences.
- **Ascites:** The majority of patients — 288 (68.9%) — did not present with ascites at enrolment. Only 24 patients (5.7%) had ascites, reflecting that ascites typically develops in late-stage liver disease. The remaining 106 patients (25.4%) had missing data, corresponding to the non-trial observational cases.

- **Hepatomegaly:** The distribution of hepatomegaly was nearly balanced among patients with available data: 152 patients (36.4%) showed no hepatomegaly, while 160 patients (38.3%) had an enlarged liver. Again, 106 patients (25.4%) had missing values. The relatively high prevalence of hepatomegaly is expected given the advanced stage of disease in much of this cohort.
- **Spider Angiomata (Spiders):** Spider angiomata were absent in 222 patients (53.1%) and present in 90 patients (21.5%), with 106 missing values (25.4%). Spider angiomata are vascular lesions associated with portal hypertension and oestrogen excess in chronic liver disease, and their presence in over one-fifth of evaluable patients reflects the systemic nature of PBC.
- **Oedema (Edema):** The large majority of patients — 354 (84.7%) — had no oedema at baseline. Slight oedema was present in 44 patients (10.5%), while severe oedema affected 20 patients (4.8%). No missing values were recorded for this variable. The progressive nature of oedema categories — from none to severe — reflects worsening fluid retention as liver function deteriorates, and this variable emerged as a significant predictor of mortality in the Cox model.
- **Histologic Disease Stage:** Disease stage at enrolment followed a distribution skewed toward advanced disease. Only 21 patients (5.0%) were classified as Stage 1, while 92 (22.0%) were Stage 2, 155 (37.1%) were Stage 3, and 144 (34.4%) were Stage 4. Six patients (1.4%) had missing stage data. The concentration of patients in Stages 3 and 4 — accounting for over 71% of the cohort — indicates that most participants had already progressed to advanced histologic disease at the time of enrolment, which is consistent with the referral patterns of a tertiary liver disease centre such as the Mayo Clinic.



3.3 Patient Follow-up

The dataset comprises 418 patients with confirmed cirrhosis who were followed longitudinally. Survival time was measured in years from enrollment to either the date of death or the last known follow-up (censoring). The primary event of interest was all-cause mortality, coded as "Dead," with surviving or lost-to-follow-up patients classified as "Censored."

Figure 1 displays the individual patient follow-up timelines for the first 50 patients in the dataset, illustrating the heterogeneity of follow-up duration and the mixture of events and censoring across the cohort.

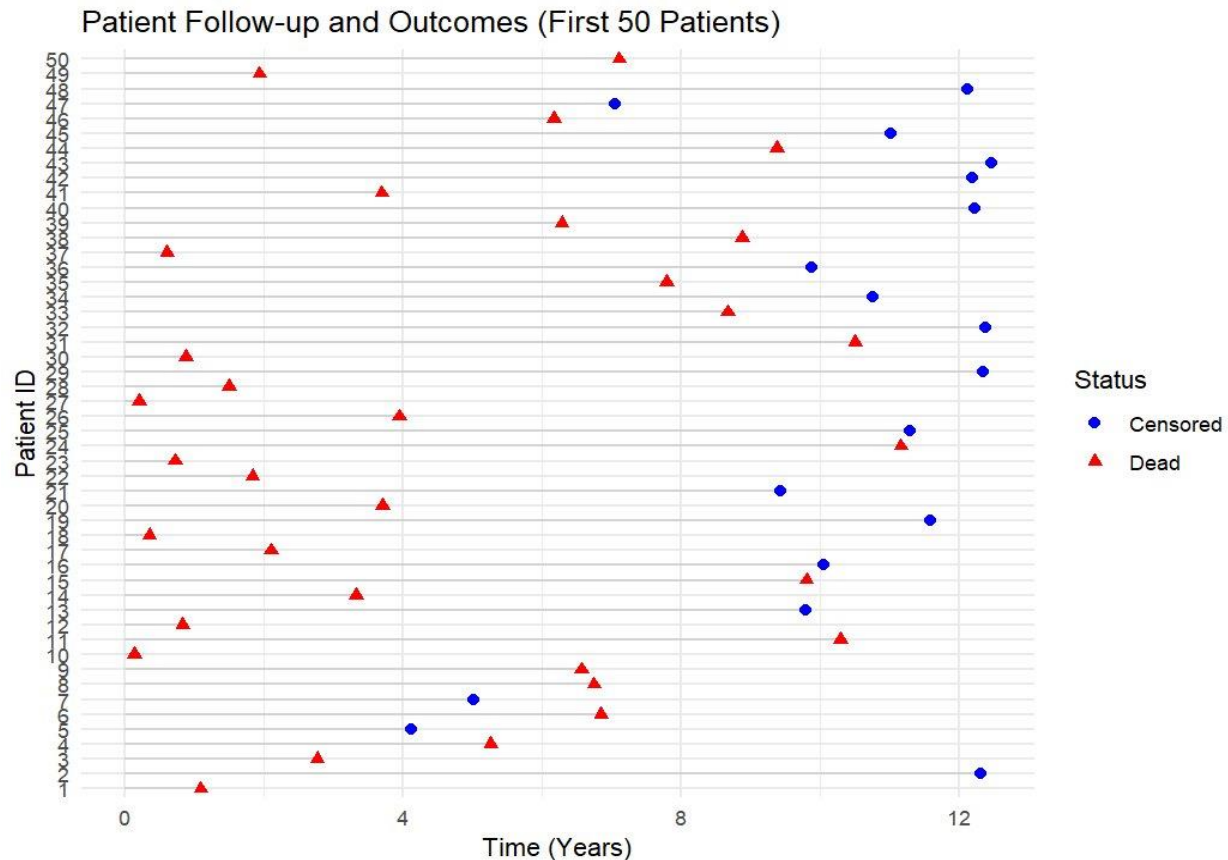


Figure 1. Patient follow-up and outcomes for the first 50 patients. Blue circles (●) represent censored observations; red triangles (▲) represent deaths. Follow-up ranges from less than 1 year to over 12 years.

As seen in Figure 1, the cohort spans a wide range of follow-up durations from under one year to over twelve years. This heterogeneity is typical of real-world longitudinal cohort studies, where patients enroll at different times and may exit observation for reasons other than the event of interest -Meier method appropriately accounts for this censored structure.

4. Kaplan-Meier Estimation and Stratified Survival Comparisons

4.1 Overall Kaplan-Meier Survival Analysis

Model Specification

The overall survival function was estimated using the Kaplan-Meier product-limit method, applied to the full cohort of 418 patients without stratification. The estimator is defined as:

$$\hat{S}(t) = \prod_{t_i \leq t} (1 - d_i / n_i)$$

where d_i is the number of deaths and n_i is the number of patients at risk at time t_i . Censoring is assumed to be non-informative (independent), which is standard in clinical cohort analyses.

Summary Statistics

Total patients: 418

Total events (deaths): 161

Censored observations: 257 (61.5%)

Median survival time: 9.3 years (95% CI: 8.46 – 10.5 years)

The median survival of 9.3 years indicates that half of the patients in this cirrhosis cohort survived beyond nine years from enrollment. The 95% confidence interval is reasonably narrow (8.46–10.5 years), suggesting adequate precision in the estimate given the available follow-up data.

Survival Probabilities at Key Time Points

Table 1 summarizes the estimated survival probabilities at clinically meaningful time horizons (1, 3, 5, and 10 years), along with associated standard errors and 95% confidence intervals.

Table 1. Overall Kaplan-Meier Survival Estimates at Selected Time Points

Time (Years)	At Risk	Events	Survival	Std. Error	95% CI
1	388	30	0.928	0.0126	0.904 – 0.953
3	312	52	0.801	0.0197	0.764 – 0.841
5	197	33	0.703	0.0236	0.658 – 0.751
10	35	41	0.442	0.0394	0.371 – 0.527

The survival estimates reveal a progressive decline: approximately 92.8% of patients survived at least 1 year; 80.1% survived to 3 years; 70.3% survived to 5 years; and only 44.2% remained alive at 10 years. The widening confidence intervals at later time points (notably at 10 years) reflect the shrinking at-risk population, where only 35 patients remained under observation.

Overall Kaplan-Meier Curve

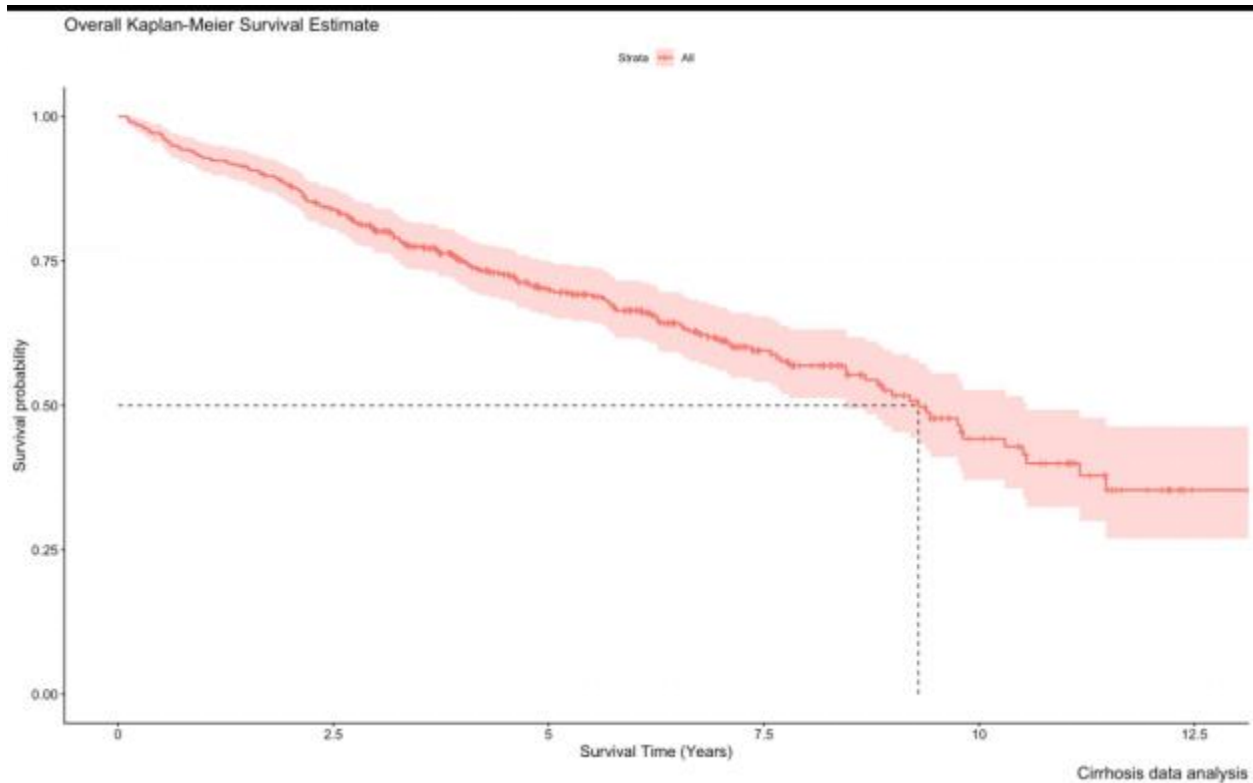


Figure 2. Overall Kaplan-Meier survival curve for the cirrhosis cohort ($n = 418$). The shaded region represents the 95% pointwise confidence interval. The dashed horizontal and vertical lines indicate the median survival time (9.3 years).

Figure 2 demonstrates a smooth, gradually declining survival curve consistent with a chronic progressive disease. The curve drops most steeply in the first five years and then decelerates, suggesting that patients who survive past the 5-year mark have a more favorable prognosis. The confidence band remains relatively narrow throughout the follow-up period, indicative of a well-powered study. The median survival line (dashed) correctly intersects the curve at approximately 9.3 years, as confirmed by the numerical output. At the end of follow-up (~13 years), the curve plateaus near 33%, reflecting a subset of long-term survivors.

4.2 Stratified Kaplan-Meier Analyses

Survival by Ascites Status

Ascites — the pathological accumulation of fluid in the peritoneal cavity — is a well-recognized and clinically important complication of cirrhosis. Its presence is a cardinal sign of hepatic decompensation and is associated with a dramatically altered prognosis. The KM curves were stratified by ascites status (Yes vs. No), with the log-rank test used to compare the two groups. The log-log confidence interval transformation was applied for improved small-sample coverage.

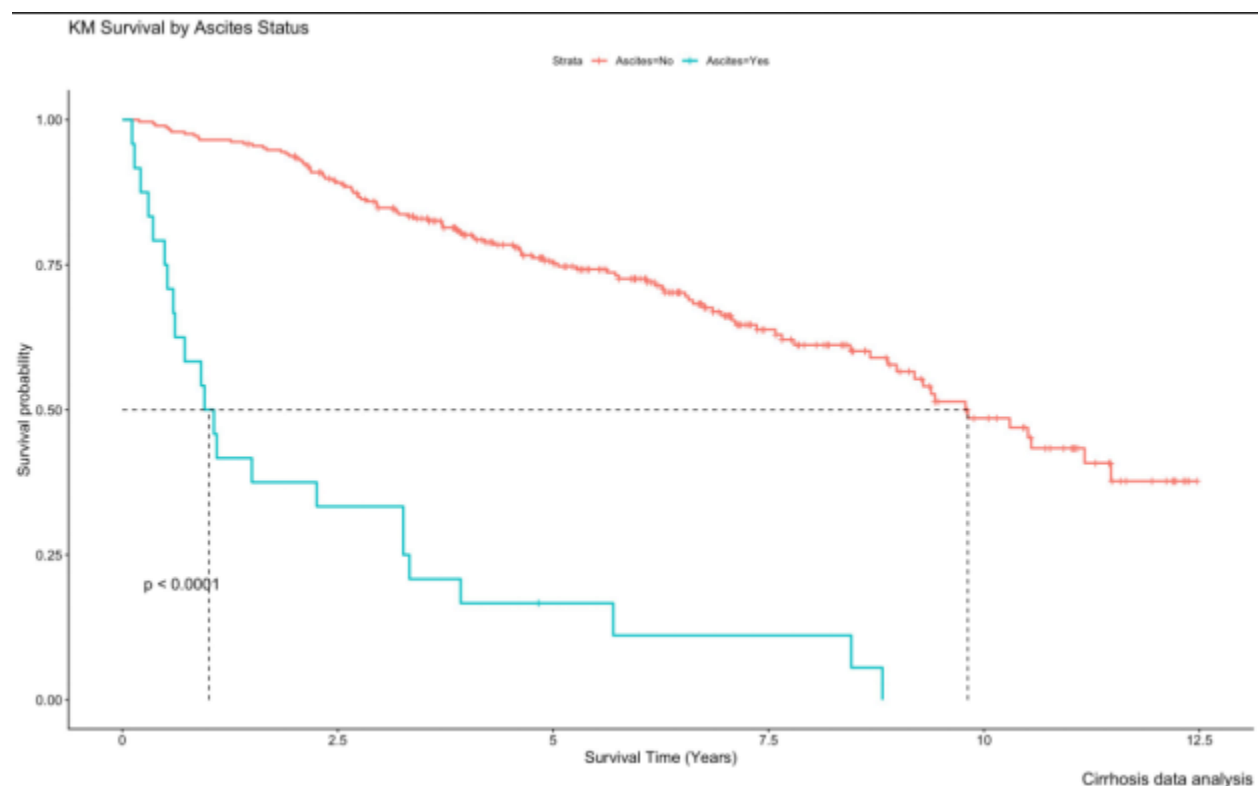


Figure 3. Kaplan-Meier survival curves stratified by ascites status. Salmon: Ascites-absent group; Teal: Ascites-present group. Median survival: ~1 year for ascites-present vs. ~10 years for ascites-absent. Log-rank $p < 0.0001$.

The survival difference between ascites groups is strikingly large and highly statistically significant ($p < 0.0001$). Patients without ascites showed a gradual decline in survival probability, with a median survival exceeding 10 years — consistent with the overall cohort estimate. In contrast, patients with ascites experienced dramatically accelerated mortality: the median survival was approximately 1 year, and survival fell below 25% by approximately 5 years, approaching 0% by 9 years.

This finding is biologically and clinically coherent. The development of ascites heralds hepatic decompensation, portal hypertension, and activation of the renin-angiotensin-aldosterone system all of which are associated with markedly worsened outcomes. The magnitude of this survival difference underscores the critical importance of ascites as a prognostic variable in cirrhosis management and should inform clinical prioritization for referral to advanced liver disease programs or transplantation evaluation.

Survival by Sex

Biological sex may influence outcomes in cirrhosis through multiple mechanisms, including differences in disease etiology (e.g., autoimmune hepatitis is more prevalent in women; alcoholic cirrhosis in men), hormonal factors affecting fibrogenesis, and differences in care-seeking behavior. The survival curves were stratified by sex (Female vs. Male).

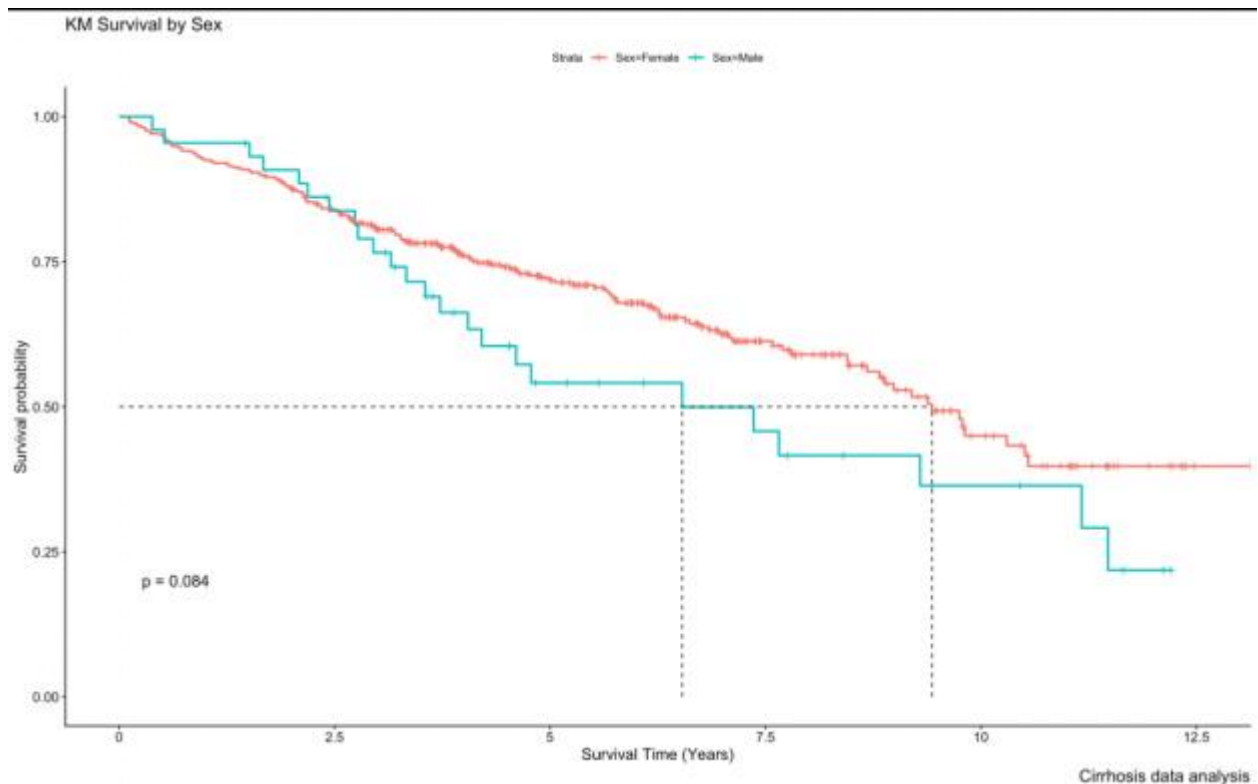


Figure 4. Kaplan-Meier survival curves stratified by sex. Salmon: Female; Teal: Male. Median survival: ~7 years (Male) vs. ~10 years (Female). Log-rank $p = 0.084$.

The KM curves by sex reveal a clinically meaningful but statistically non-significant trend ($p = 0.084$). Male patients had a shorter median survival of approximately 7 years compared to approximately 10 years in females. The two curves overlap substantially in the first 2 years and diverge progressively thereafter, with males exhibiting a steeper decline through mid-follow-up. By 12 years, the terminal survival estimates differ appreciably: females plateau near 39% while males plateau near 24%.

The p-value of 0.084 falls short of the conventional 0.05 threshold, meaning that by the standard frequentist criterion, sex alone does not constitute a statistically significant prognostic factor in this cohort. However, this result should be interpreted cautiously. The observed trend is clinically plausible and may represent a real biological effect that the study was underpowered to detect particularly given potential confounding by disease etiology and severity at baseline. Sex should be considered as a covariate in multivariable Cox regression analyses rather than dismissed as a prognostic factor based solely on this univariable KM comparison.

Survival by Hepatomegaly

Hepatomegaly enlargement of the liver can occur in cirrhosis due to inflammatory infiltration, fatty change, vascular congestion, or regenerative nodule formation. Its presence may reflect either compensated or decompensated disease, depending on the underlying etiology. The KM analysis stratified patients by the presence or absence of hepatomegaly.

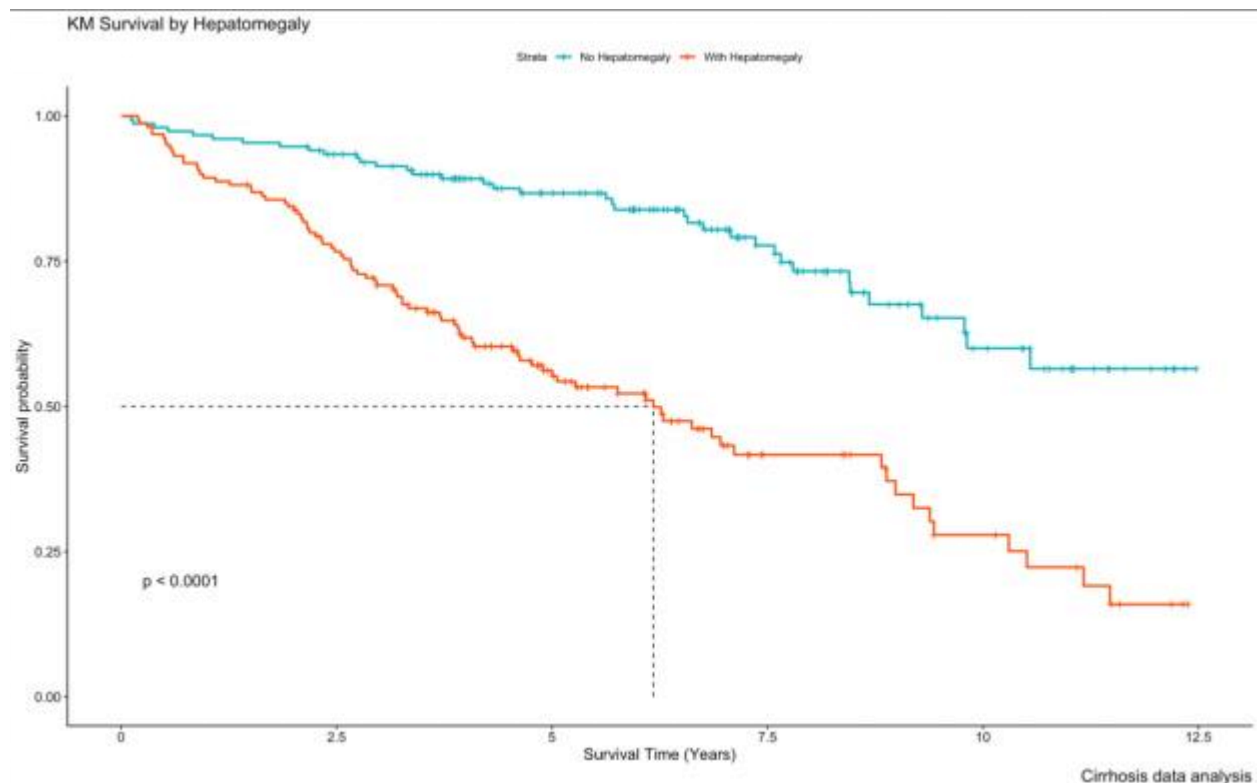


Figure 5. Kaplan-Meier survival curves stratified by hepatomegaly status. Teal: No Hepatomegaly; Orange: With Hepatomegaly. Median survival: ~5.5 years (hepatomegaly) vs. not reached in follow-up period (no hepatomegaly). Log-rank $p < 0.0001$.

The stratification by hepatomegaly reveals the most pronounced separation observed across all subgroup analyses. Patients without hepatomegaly maintained high survival probabilities well into the follow-up period exceeding 85% at 5 years and remaining above 55% even at 12 years, with

the median survival not reached within the observation window. Conversely, patients with hepatomegaly showed substantially worse outcomes: median survival was approximately 5.5 years, and survival declined to approximately 15-20% by 12 years.

The highly significant p-value (< 0.0001) confirms that hepatomegaly is a strong prognostic indicator in this cohort. This finding may reflect that hepatomegaly in cirrhotic patients is a marker of more advanced or active hepatic pathology, including portal hypertension, active inflammation, or infiltrative disease. Clinically, the presence of hepatomegaly should trigger a comprehensive diagnostic workup and may indicate a need for more aggressive management or expedited transplantation evaluation.

4.3 Comparative Summary of Stratified Analyses

Table 2 provides a structured summary of the key findings from the three stratified Kaplan-Meier analyses, enabling direct comparison across prognostic variables.

Table 2. Summary of Stratified Kaplan-Meier Analyses

Variable	Group	Median Survival	Log-rank p	Interpretation
Ascites	No	~10 years	< 0.0001	Strong prognostic factor
Sex	Yes	~1 year	0.084	Trend; not significant
	Female	~10 years		
	Male	~7 years		
Hepatomegaly	Absent	Not reached	< 0.0001	Strong prognostic factor
	Present	~5.5 years		

Both ascites and hepatomegaly emerge as highly statistically significant independent predictors of survival ($p < 0.0001$), with ascites producing the most dramatic absolute difference in median survival (~9 years between groups). Sex shows a clinically suggestive but non-significant trend at conventional thresholds.

4.4 Discussion

This survival analysis of a cirrhosis cohort provides several clinically important insights. The overall median survival of 9.3 years reflects the chronic nature of cirrhosis when managed in a clinical setting patients may survive for many years with appropriate monitoring and intervention. However, survival varies enormously based on disease complications.

The finding that ascites reduces median survival from approximately 10 years to approximately 1 year is perhaps the most clinically actionable result. Ascites is a marker of hepatic decompensation and should be treated as a medical emergency requiring urgent optimization and consideration of transplantation. This result is consistent with the natural history literature on cirrhosis, where decompensation events (ascites, encephalopathy, variceal bleeding) portend a 1-year mortality exceeding 50%.

The hepatomegaly finding is equally compelling: patients without hepatomegaly appeared to have excellent long-term survival, suggesting a distinct disease trajectory. This may reflect a subgroup with early or compensated cirrhosis, or a primary biliary cirrhosis population where hepatomegaly is absent despite significant disease. Further clinical stratification by disease etiology and Child-Pugh/MELD scores would be needed to fully interpret this finding.

The sex analysis, while not meeting the conventional significance threshold, aligns with known epidemiological data suggesting that women with cirrhosis particularly those with autoimmune or cholestatic etiologies may have somewhat more favorable outcomes than men. This warrants further investigation in a multivariable framework, controlling for confounders.

This Kaplan-Meier survival analysis of 418 cirrhosis patients yields the following principal conclusions:

- The overall median survival was 9.3 years (95% CI: 8.46–10.5), with 1-, 3-, 5-, and 10-year survival probabilities of 92.8%, 80.1%, 70.3%, and 44.2%, respectively.
- Ascites is a potent negative prognostic factor, reducing median survival by approximately 9 years ($p < 0.0001$). Patients with ascites should be prioritized for advanced interventions.
- Hepatomegaly is associated with markedly worse survival ($p < 0.0001$), with a median survival of ~5.5 years compared to a median not reached in hepatomegaly-absent patients.
- Sex shows a trend toward worse survival in males (median ~7 vs. ~10 years), but this did not reach statistical significance ($p = 0.084$) in this cohort.

5. Cox Proportional Hazards Regression Analysis

A Cox proportional hazards regression model was used to identify the factors associated with mortality risk among patients with primary biliary cirrhosis. This model examines how different clinical and demographic variables affect patients' survival time while taking censored observations into account.

In this study, survival time was measured in years using the variable Years, which was created by converting the follow-up time (N_Days) from days into years. The event variable was Status, where patients classified as "Dead" were considered as observed events, while patients classified as "Censored" were considered censored observations.

This analysis aimed to determine which patient characteristics had a significant effect on the hazard of death during the follow-up period.

5.1 Full Cox Regression Model

The full Cox proportional hazards regression model was fitted using all available demographic and clinical variables to evaluate their association with mortality risk among patients with primary biliary cirrhosis.

5.2 The fitted Cox model was:

$$h(t) = h_0(t) \exp (\beta_1 Age + \beta_2 Sex + \beta_3 Drug + \beta_4 Ascites + \beta_5 Hepatomegaly + \beta_6 Spiders + \beta_7 Edema + \beta_8 Bilirubin + \beta_9 Cholesterol + \beta_{10} Albumin + \beta_{11} Copper + \beta_{12} Alk_Phos + \beta_{13} SGOT + \beta_{14} Tryglicerides + \beta_{15} Platelets + \beta_{16} Prothrombin + \beta_{17} Stage)$$

where:

- $h(t)$ represents the hazard function at time t
- $h_0(t)$ represents the baseline hazard function
- β_i represents the regression coefficient associated with each covariate

5.3 Significant and In Significant Variables in the Full Cox Model

Variable	Hazard Ratio (HR)	P-value	Significance
Age	1.000	0.013	Significant
Sex	1.456	0.227	In significant
Drug	0.838	0.420	In significant
Ascites	1.001	0.999	In significant
Hepatomegaly	1.058	0.826	In significant
Spiders	1.073	0.778	In significant
Edema (Slight)	1.276	0.468	In significant
Edema (Severe)	3.146	0.006	Significant
Bilirubin	1.083	0.002	Significant
Cholesterol	1.000	0.297	In significant
Albumin	0.473	0.016	Significant
Copper	1.002	0.037	Significant
Alk_Phos	1.000	0.990	In significant
SGOT	1.004	0.062	In significant
Tryglicerides	0.999	0.718	In significant
Platelets	1.001	0.479	In significant
Prothrombin	1.318	0.018	Significant
Stage 2	4.081	0.193	In significant
Stage 3	5.380	0.110	In significant
Stage 4	8.325	0.047	Significant

5.4 Results

In the full Cox model, several variables had hazard ratios greater than 1, indicating increased mortality risk. The highest hazard ratios were observed for Stage 4 (HR = 8.325), Stage 3 (HR = 5.380), Stage 2 (HR = 4.081), and Severe Edema (HR = 3.146). However, only Severe Edema and Stage 4 were statistically significant among these variables.

Other significant predictors associated with increased mortality risk included Bilirubin (HR = 1.083), Copper (HR = 1.002), and Prothrombin (HR = 1.318). In contrast, Albumin showed a protective effect (HR = 0.473), indicating better survival probability with higher albumin levels.

Although the full model showed good predictive performance (Concordance = 0.848), many variables were statistically non-significant. Therefore, a reduced Cox model was later fitted using the most significant and clinically relevant predictors to improve model simplicity and interpretability.

5.5 Reduced Cox Regression Model

The reduced Cox proportional hazards regression model was fitted using the most significant and clinically relevant variables identified from the full model to evaluate their association with mortality risk among patients with primary biliary cirrhosis.

5.6 The fitted reduced Cox model was

$$h(t) = h_0(t) \exp (\beta_1 \text{Age} + \beta_2 \text{Edema} + \beta_3 \text{Bilirubin} + \beta_4 \text{Albumin} + \beta_5 \text{Copper} + \beta_6 \text{Prothrombin} + \beta_7 \text{Stage})$$

5.7 Significant and In Significant Variables in the Reduced Cox Model

Variable	Hazard Ratio (HR)	P-value	Significance
Age	1.000	0.003	Significant
Edema (Slight)	1.057	0.847	In significant
Edema (Severe)	2.480	0.004	Significant
Bilirubin	1.108	0.001	Significant
Albumin	0.419	0.001	Significant
Copper	1.003	0.001	Significant
Prothrombin	1.348	0.003	Significant
Stage 2	5.792	0.100	In significant
Stage 3	7.201	0.058	In significant
Stage 4	9.543	0.030	Significant

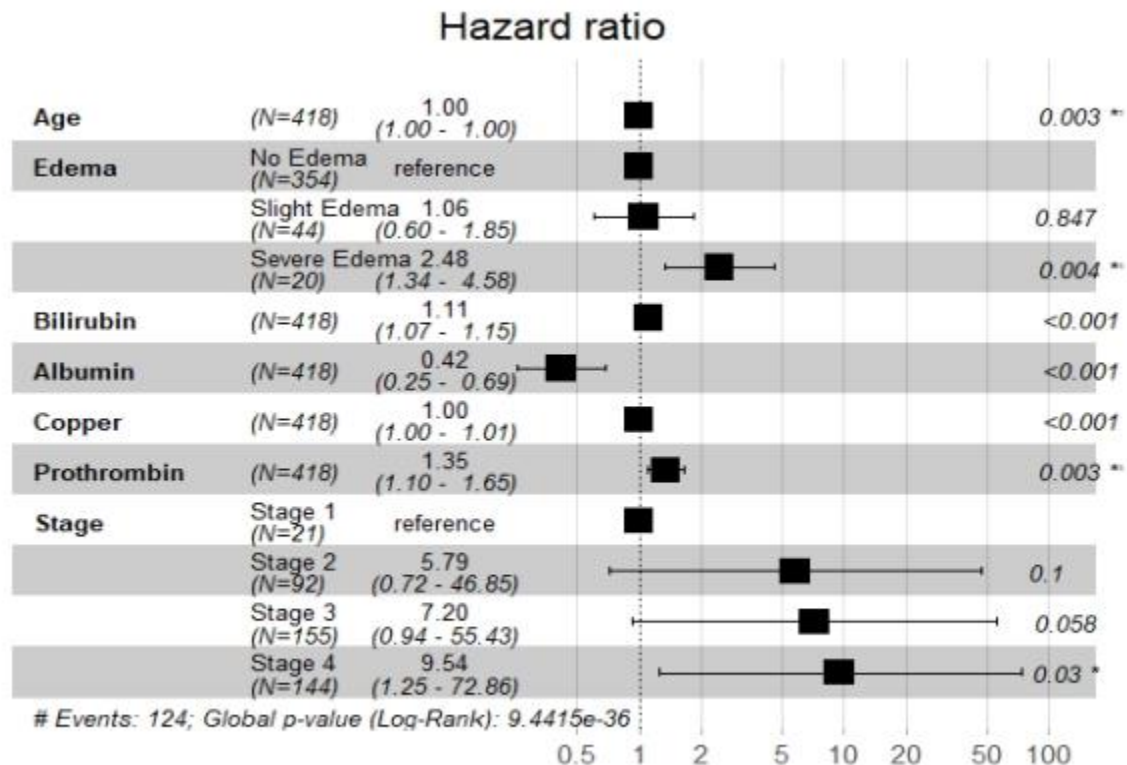
5.8 Results

In the reduced Cox model, the highest hazard ratios were observed for Stage 4 (HR = 9.543), Stage 3 (HR = 7.201), Stage 2 (HR = 5.792), and Severe Edema (HR = 2.480), indicating increased mortality risk with disease severity. Among these variables, Severe Edema and Stage 4 remained statistically significant predictors of increased mortality risk.

Additional significant risk factors included Bilirubin (HR = 1.108), Copper (HR = 1.003), and Prothrombin (HR = 1.348), all associated with higher hazard of death. In contrast, Albumin remained the main protective factor (HR = 0.419), suggesting improved survival probability and reduced mortality risk with higher albumin levels.

Compared with the full Cox model, the reduced model showed slightly improved predictive performance (Concordance = 0.849) while using fewer variables. In addition, the likelihood ratio, Wald, and score tests were all highly significant, indicating that the reduced model remained statistically strong and clinically meaningful.

5.9 Forest Plot of the Reduced Cox Proportional Hazards Model



The forest plot summarizes the hazard ratios and 95% confidence intervals for the variables included in the reduced Cox proportional hazards model.

Variables with hazard ratios greater than 1 were associated with increased mortality risk, particularly Stage 4 disease (HR = 9.54), Severe Edema (HR = 2.48), Bilirubin (HR = 1.11), and Prothrombin (HR = 1.35). These variables were associated with poorer survival outcomes.

In contrast, Albumin showed a protective effect with HR = 0.42, indicating lower mortality risk and better survival probability with higher albumin levels.

The plot also shows that several variables, including Severe Edema, Bilirubin, Albumin, Copper, Prothrombin, and Stage 4, were statistically significant because their confidence intervals did not cross the reference line at HR = 1.

6. Proportional Hazards Assumption

The proportional hazards assumption was formally assessed for the reduced Cox model using the Schoenfeld residuals test via `cox.zph()`.

For each individual variable:

- **H₀:** The hazard ratio for the predictor is constant over time, the proportional hazards assumption holds.
- **H₁:** The hazard ratio for the predictor changes over time, the proportional hazards assumption is violated.

For the global test:

- **H₀:** The hazard ratios for all predictors in the model are constant over time, the proportional hazards assumption holds globally across the entire model.
- **H₁:** At least one predictor in the model has a hazard ratio that changes over time, the proportional hazards assumption is violated for at least one variable.

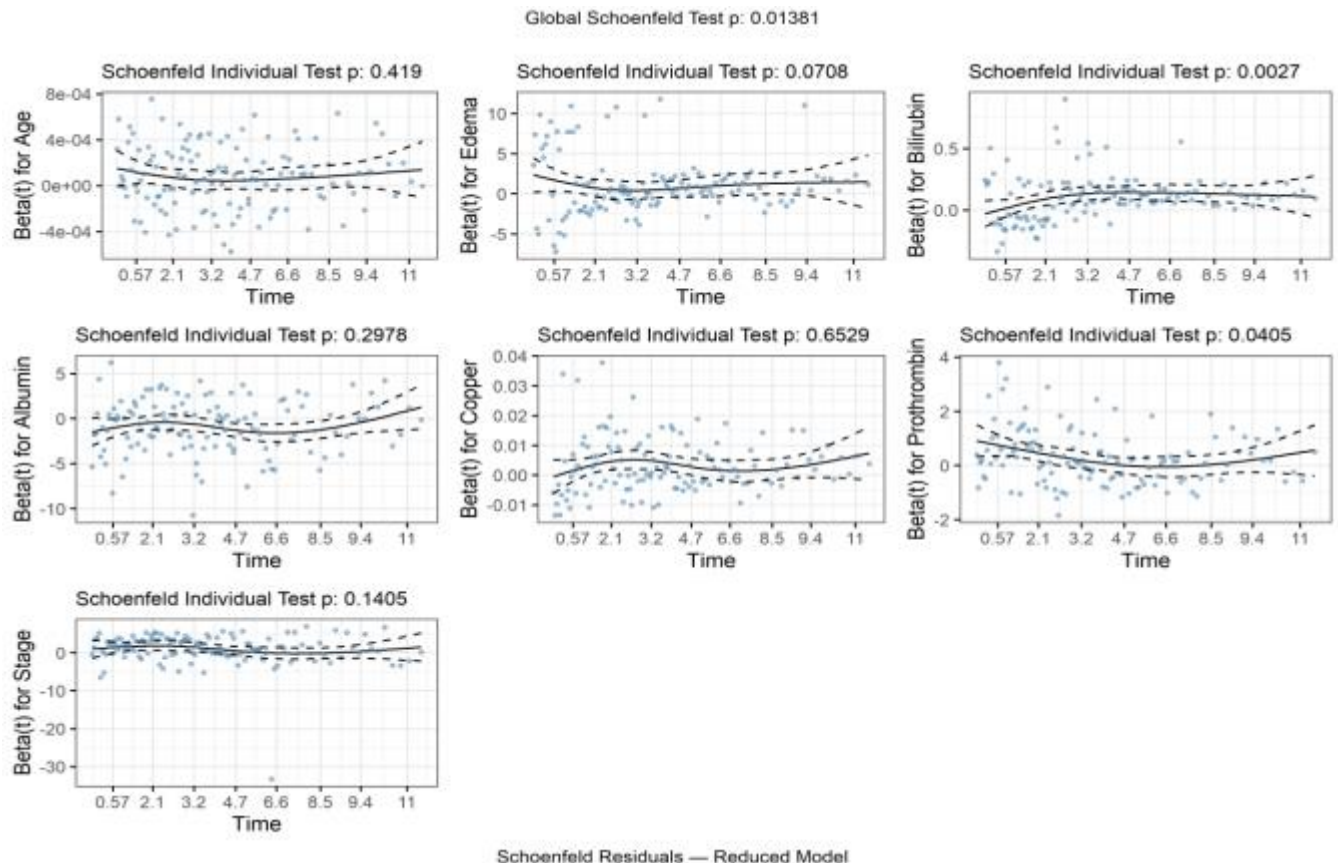
Variable-level results:

- **Age ($\chi^2 = 0.653$, $df = 1$, $p = 0.419$)** The PH assumption is satisfied for Age. The hazard ratio associated with age remained stable throughout the follow-up period, confirming that the effect of age on mortality risk did not change over time.
- **Edema ($\chi^2 = 5.295$, $df = 2$, $p = 0.071$)** The PH assumption is satisfied for Edema, though the p-value is borderline (just above 0.05). This suggests a marginal trend toward non-proportionality that does not reach statistical significance. The result should be noted with caution but does not require corrective action.
- **Bilirubin ($\chi^2 = 9.021$, $df = 1$, $p = 0.003$)** The PH assumption is **violated** for Bilirubin. This is the most significant violation in the model, indicating that the effect of serum bilirubin on mortality risk is not constant over time. Clinically, this is plausible — bilirubin reflects dynamic disease progression in PBC, and its prognostic impact likely diminishes or intensifies as the disease advances.
- **Albumin ($\chi^2 = 1.084$, $df = 1$, $p = 0.298$)** The PH assumption is satisfied for Albumin. The protective effect of albumin on survival remained proportional and stable across the entire follow-up period.
- **Copper ($\chi^2 = 0.202$, $df = 1$, $p = 0.653$)** The PH assumption is satisfied for Copper, with the highest p-value in the model. There is no evidence whatsoever that the effect of urinary copper on mortality changes over time.
- **Prothrombin ($\chi^2 = 4.199$, $df = 1$, $p = 0.041$)** The PH assumption is **violated** for Prothrombin time. Although the violation is less severe than that of Bilirubin, the p-value falls below the 0.05 threshold, indicating that the hazard ratio for prothrombin time is not constant over

follow-up. This is clinically interpretable — prothrombin time may become a stronger predictor of imminent death as liver synthetic function progressively fails in late-stage disease.

- **Stage ($\chi^2 = 5.469$, $df = 3$, $p = 0.141$)** The PH assumption is satisfied for histologic Stage. Despite Stage 4 being a strong predictor of mortality, its proportional effect on the hazard remained consistent over time.

6.1 Schoenfeld Residuals Plots



The visual inspection of the Schoenfeld residual plots is consistent with the formal statistical test results. Five variables: Age, Edema, Albumin, Copper, and Stage display flat or near-flat smoothed lines confirming proportional hazards. Two variables: Bilirubin and Prothrombin display clear downward slopes over time, visually confirming their statistically significant violations. Both violating variables show the same directional pattern: a stronger early effect that diminishes over follow-up, which is clinically interpretable in the context of progressive liver disease where early biochemical deterioration may be most predictive of near-term mortality.

6.2 Stratification (Stratified Cox Proportional Hazards Model)

Following the detection of proportional hazards violations in Bilirubin and Prothrombin time, a stratified Cox proportional hazards model was fitted as the final analytical model. The model was estimated on **310 patients with 124 observed deaths**, with 108 observations excluded due to missing values. The model stratified by Bilirubin group (Normal, Elevated, High) and Prothrombin group (Normal, Prolonged), and estimated hazard ratios for four remaining covariates: Age, Edema, Albumin, and Copper.

6.3 Overall Model Significance

The model was highly statistically significant across all three global tests — Likelihood Ratio ($p=1\times10^{-11}$), Wald ($p=9\times10^{-12}$), and Score log-rank ($p=4\times10^{-13}$) — confirming that the four covariates together explain a large and non-random portion of the variation in survival time among cirrhosis patients.

Age was a statistically significant predictor of mortality ($p=0.0006$). Although the hazard ratio per day is negligibly small ($HR=1.000$), scaled to years this corresponds to approximately a **3% increase in the hazard of death per additional year of age**. This confirms that older patients carry a consistently elevated mortality risk, independent of their clinical and biochemical status, likely reflecting reduced physiological reserve and greater comorbidity burden.

Severe Edema was the strongest clinical predictor in the model. Patients with severe edema had **2.928 times the hazard of death** compared to patients with no edema (95% CI: 1.601–5.355, $p=0.0005$). This is a highly significant and clinically meaningful finding, as severe edema reflects advanced portal hypertension, sodium retention, and critically impaired hepatic synthetic function all hallmarks of end-stage liver disease. In contrast, **slight edema** showed a non-significant elevation in risk ($HR=1.552$, 95% CI: 0.886–2.720, $p=0.124$), suggesting that mild fluid retention alone does not independently confer a statistically detectable survival disadvantage once other factors are controlled.

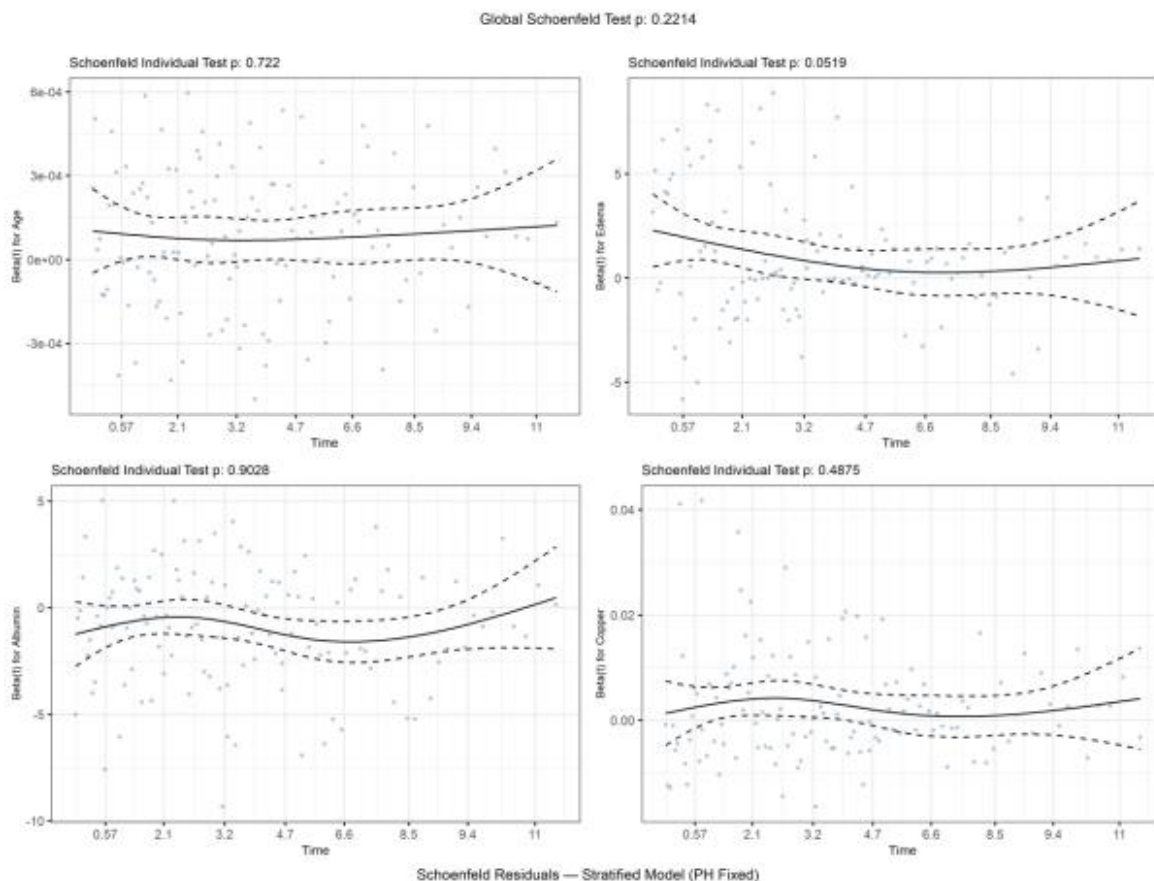
Albumin was the strongest **protective** factor in the model ($p=0.0002$). Each 1 g/dL increase in serum albumin was associated with a **58.9% reduction in the hazard of death** ($HR=0.411$, 95% CI: 0.256–0.660). Since albumin is synthesized exclusively by the liver, its concentration is a direct and sensitive marker of remaining hepatic synthetic capacity. This result reinforces albumin as the single most important biochemical indicator of prognosis in this cohort, consistent with its central role in established scoring systems such as Child-Pugh and MELD.

Copper remained a statistically significant independent predictor of mortality ($HR=1.003$ per $\mu\text{g/day}$, 95% CI: 1.001–1.005, $p=0.007$). While the per-unit effect is modest, the wide range of urinary copper values in this cohort (4 to 588 $\mu\text{g/day}$) means the cumulative impact across patients is clinically relevant. Elevated urinary copper in PBC reflects impaired biliary copper excretion

due to bile duct destruction, and its association with poorer survival is consistent with the severity of underlying cholestatic injury.

Re-testing the proportional hazards assumption on the stratified model confirmed that all four remaining covariates fully satisfied the assumption: Age ($p=0.722$), Edema ($p=0.052$), Albumin ($p=0.903$), and Copper ($p=0.488$). Critically, the global test was non-significant ($\chi^2=6.99$, $df=5$, $p=0.221$), confirming that the stratified model is free from proportionality violations and that its hazard ratio estimates are valid, stable, and interpretable across the entire follow-up period.

6.4 Schoenfeld Residuals Plots for the stratified cox model

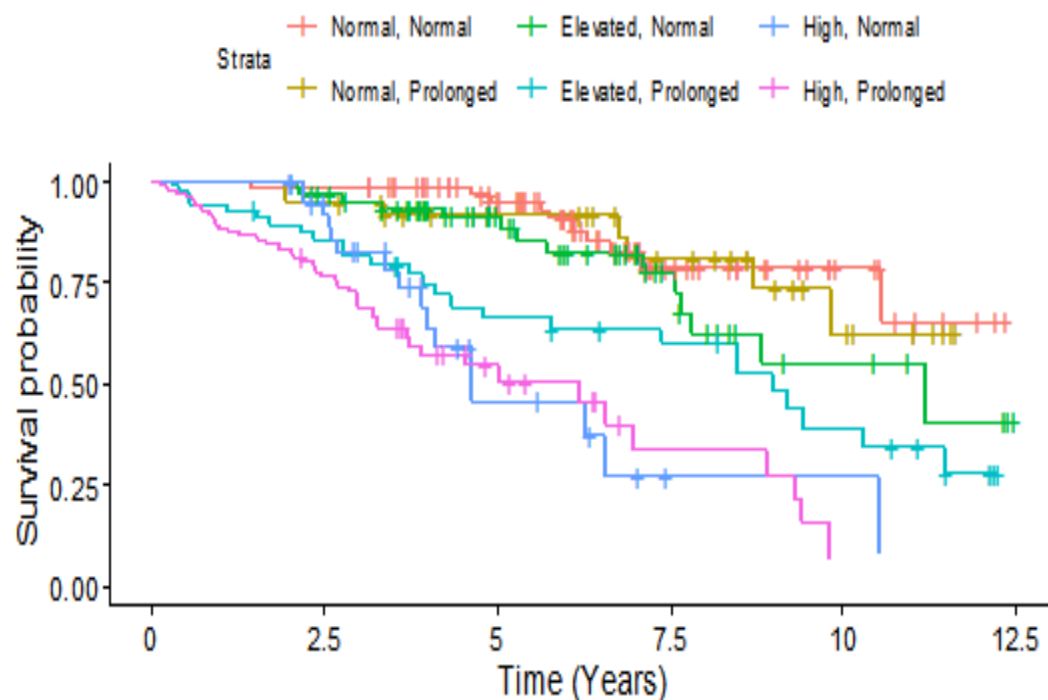


The Schoenfeld residuals plot for the stratified Cox model provides clear visual and statistical confirmation that the proportional hazards assumption is satisfied for all four covariates Age, Edema, Albumin, and Copper both individually and globally. All smooth lines are approximately flat, all residuals are randomly scattered with no directional trend, all individual p-values exceed 0.05, and the global p-value of 0.221 confirms no overall violation. By stratifying on Bilirubin and Prothrombin time, the earlier violations detected in the reduced model have been completely resolved, and the hazard ratio estimates from the stratified model can be interpreted with full methodological confidence.

6.5 Stratified Cox Model Survival Curves

Stratified Cox Model Survival Curves presents the adjusted survival curves estimated by the stratified Cox model, displayed separately for each combination of Bilirubin group (Normal, Elevated, High) and Prothrombin group (Normal, Prolonged), yielding six distinct strata. The curves demonstrate a clear and clinically interpretable gradient of survival: patients with Normal Bilirubin and Normal Prothrombin time experienced the best long-term survival (approximately 65–70% at 12.5 years), while patients with High Bilirubin and Prolonged Prothrombin time faced the most severe prognosis, with survival falling to near zero by Year 10–11. Within each Bilirubin category, prolongation of prothrombin time consistently shifted the survival curve downward, confirming the independent additive prognostic burden of coagulopathy. The non-parallel and crossing nature of the six curves across follow-up time provides direct visual confirmation that the effects of Bilirubin and Prothrombin on the hazard of death are not constant over time, validating the methodological decision to stratify by these variables rather than include them as standard Cox covariates.

Stratified Cox Model Survival Curves



7. Conclusion

This study applied several survival analysis techniques to evaluate mortality risk among patients with Primary Biliary Cirrhosis using the Mayo Clinic PBC dataset. The Kaplan–Meier analysis showed that survival gradually decreased over time, with an overall median survival of approximately 9.3 years. Clear differences in survival were observed between patient groups, particularly for patients with ascites and hepatomegaly, both of which were associated with markedly poorer outcomes.

The Cox proportional hazards models identified a number of important predictors of mortality. Higher bilirubin, copper, prothrombin time, severe edema, advanced disease stage, and older age were associated with increased risk of death, while higher albumin levels were associated with better survival. Among these factors, albumin appeared to be the strongest protective factor, reflecting the importance of preserved liver function in improving prognosis.

Testing the proportional hazards assumption revealed that bilirubin and prothrombin time did not satisfy the assumption over follow-up time. To address this issue, a stratified Cox model was fitted, which successfully resolved the proportionality violations and produced stable and interpretable hazard estimates. The final stratified model confirmed that age, severe edema, low albumin, and elevated copper remained significant independent predictors of mortality.

Overall, the findings of this study are clinically consistent with the known progression of primary biliary cirrhosis and highlight the major role of liver function indicators and disease severity in determining patient survival. The analysis also demonstrates the importance of validating model assumptions in survival analysis to ensure reliable statistical conclusions.

8. References

1. Kleinbaum, D. G., & Klein, M. (2012). Survival Analysis: A Self-Learning Text (3rd ed.). Springer.

<https://link.springer.com/book/10.1007/978-1-4419-6646-9>

2. Fleming, T. R., & Harrington, D. P. (1991). Counting Processes and Survival Analysis. Wiley.

<https://onlinelibrary.wiley.com/doi/book/10.1002/9781118150653>

3. Dickson, E., Grambsch, P., Fleming, T., Fisher, L., & Langworthy, A. (1989). Cirrhosis Patient Survival Prediction Dataset. UCI Machine Learning

<https://archive.ics.uci.edu/dataset/878/cirrhosis+patient+survival+prediction+dataset-1>